Run : **Groupe : GIE** **Date :** / / 20__

Surnames and first names of the members of the Working Group:

-	-
-	-
-	-
-	-

List in the first table below all the problems encountered and assess their criticality with respect to meeting commitments.

Echelle proposée :

O : Probability of Occurrence: :	1 – Very Low	⟷	4 – Very High
I : Impact on Commitment Compliance (Cost / Quality / Deadline)	1 – Very Low Impact	⟷	4 – Very High Impact 5 – Unacceptable Impact
C : Criticality	C = O * I	Range of values :	1 à 20

List in the second table below all the improvements selected to be implemented for the next run; assign the responsible persons for carrying out each task.

© 2019 F. ROSIN

N°	List of problems encountered	O	I	C
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

N°	List of selected improvements Action plan	Task Owner
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		

Work description

Part 1 — Pair work:

Q1-1

Based on the description of the TPS house (Toyota Production System), explain how decision-making is a central element of a Lean production system and, more generally, of Lean culture (you may want to look for additional information about the *Toyota Way*).

Q1-2

You are then asked to link:

- the different stages of the decision-making process (see Rosin et al. (2021): "Impact of Industry 4.0 on decision-making in an operational context"), and
- the different concepts associated with a Lean production system (see the TPS house).

The result may be presented as a matrix (e.g., rows: the various Lean concepts; columns: the different stages of a decision-making process + others). Each cell can then include a short, reasoned comment.

Q1-3

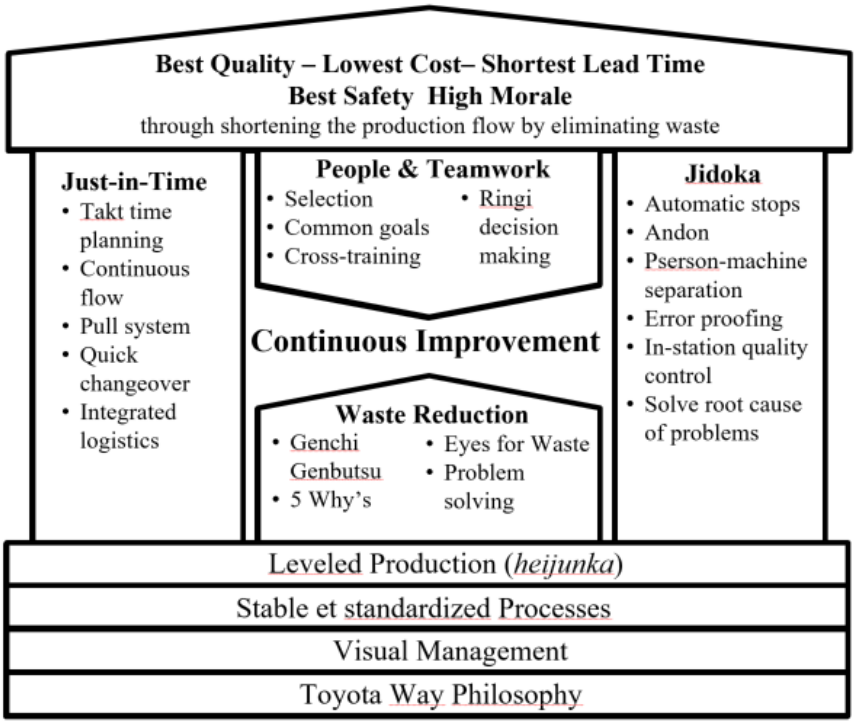
Relying on the typology proposed by Romero et al. (2016), which type(s) of Operator 4.0 do you think are most closely linked to the strengthening of the decision-making process by new technologies?

Part 2 — Individual work:

Each of you must choose one of the problems identified at the end of the simulated run and carry out a problem-solving exercise following the 8-step methodology of Toyota Business Practices (TBP) (see description below). Then propose one or more technologies that you think could help manage similar problems.

Explain which step(s) of the problem-solving process these technologies can facilitate and how.

TPS house (Toyota Production System)



Liker, J. K. (2004) The Toyota way: 14 management principles from the world's greatest manufacturer, McGraw-Hill: New York.

The Toyota Production System (TPS) is often visualized as a house, where the roof represents the goals, the pillars represent the main operational elements, and the foundation ensures stability.

The Goals (The Roof)
The top of the house represents the primary objectives of the system :

- Best Quality, Lowest Cost, Shortest Lead Time: These are the goals stated on the roof of the TPS House diagram
- The organization often follows QCDSM (Quality, Cost, Delivery, Safety, and Morale)

The Pillars
The two main operating concepts that support the system are **Just-in-Time** and **Jidoka**

Just-in-Time (JIT): This is the philosophy of delivering **the right items at the right time in the right amounts**. It is a set of principles, tools, and techniques that allow a company to produce and deliver products in small quantities, with short lead times, to meet specific customer needs

Takt time planning
Takt time is defined as the rate of customer demand—the pace at which the customer is buying product. Takt time planning is the strategic use of Takt time to design manufacturing or service processes, made possible by Heijunka (the process of leveling the production mix and volume). This integrated approach allows the organization to achieve high productivity, standardized work, and responsiveness to demand without stressing people or equipment.

Continuous Flow

Continuous flow is the ideal of Just-in-Time (JIT) manufacturing, aiming for the product to move through value-added processes without interruption. In its most "pure" form, One-Piece Flow involves producing a single unit at a time at a pace determined by customer demand, called the takt time.
The most critical function of continuous flow in the TPS is that it brings problems to the surface.
Continuous flow connects processes and people with each other, drastically reducing intermediate inventories (Work-In-Process or WIP).

Pull Systems
The basic principle of JIT is material replenishment initiated by consumption. It means providing the downline customers with what they want, when they want it, and in the amount they want, thus minimizing work in process and inventory.
→ **Kanban**
The tool/signal system for managing and ensuring the flow and production of materials in a just-in-time production system. It is the Japanese word for "card," "ticket," or "sign" used to signal when parts are needed.

Quick Changeover
Techniques to minimize the time required to change equipment or a production line from making one product to another. This allows for smaller batches and mixed-model production, which are essential for JIT and Heijunka.

Integrated logistics
This concept refers to managing the entire flow of materials and information across the extended enterprise, from raw materials through suppliers to the final customer, as a cohesive, single system focused on Lean principles

Jidoka (Built-in Quality / Autonomation): This means never letting a defect pass into the next station and freeing people from machines. It involves building into equipment the capability of detecting problems and stopping itself. Jidoka is automation with a human touch.

Automatic Stops
Building into the equipment the capability of detecting problems and stopping itself. This concept originated with Sakichi Toyoda's loom invention that would automatically stop a loom whenever a thread broke.

Andon
A visual control device in a production area that alerts workers to defects, equipment abnormalities, or other problems using signals such as lights or audible alarms. It refers to the light signal for help, and every team member has the responsibility to stop the line every time they see something out of standard.

Person-Machine Separation
Designing operations and equipment so your workers are not tied to machines and are free to perform value-added work. This process is key to freeing people from machines.

Error Proofing
Known as Poka-yoke (a mechanism that prevents an operator from making an error). It is synonymous with the philosophy of "mistake proofing" and is essential to creating continuous flow.

In-station Quality Control
The practice of preventing problems from being passed down the line, which is more effective and less costly than inspecting and repairing defects later. It means having a culture of stopping or slowing down to get quality right the first time.

Solve Root Cause of Problems (5 Why's)
A problem-solving technique involving asking "Why?" five times to continually dig past the surface appearance (source) and identify the root cause of the problem. This rigorous analysis is central to continuous organizational learning.

The Core

Continuous Improvement (Kaizen): This defines Toyota's basic approach to doing business. It is the process of making incremental improvements (no matter how small) and eliminating all waste (muda).

Waste reduction
L'apprentissage organisationnel, qui est atteint en résolvant sans relâche les problèmes de fond. L'identification des causes profondes de ces problèmes (via le 5 Why's et le Problem Solving) et la compréhension de la situation réelle (via le Genchi Genbutsu) permettent d'éliminer le gaspillage (Muda) et les irrégularités (Mura) qui causent une surcharge (Muri)

Genchi Genbutsu
Go and see for yourself to thoroughly understand the situation. It literally means going to the actual location (Genchi) to observe and verify the actual materials or products (Genbutsu) firsthand. This approach ensures that decisions are based on personally verified data and facts, not just theory.

5 Why's
A rigorous problem-solving technique involving asking "Why?" five times to continually dig past the surface appearance (source) and identify the root cause of the problem. This analysis is central to continuous organizational learning and developing effective countermeasures.

Eye for Waste
The trained mindset and cultural ability to see waste (Muda) by observing processes and identifying non-value-added activities. Taiichi Ohno defined the goal of Lean as reducing the time line by removing the non-value-added wastes.

Problem Solving
The continuous process of identifying root causes of problems and preventing them from occurring. It is the focus of Toyota's continuous learning system and is often done through a systematic, eight step process. At Toyota, problem solving is considered 80% thinking and 20% tools.

People & Teamwork
The system centers around the people and the culture of continuous learning

Selection
The practice of putting tremendous effort into finding and screening prospective employees. Toyota focuses on selecting individuals whose values and personal characteristics can be molded into the Toyota Way philosophy.

Common Goals
The leadership effort to motivate and engage large numbers of people to work together toward a common goal. The philosophical mission is to align the whole organization toward a common purpose that is bigger than making money. This is formally achieved through Policy Deployment (**Hoshin Kanri**), which cascades objectives down to all employees to align continuous improvement efforts.

Cross Training
The necessary practice of training individuals to be multi-skilled to work across different manufacturing processes. This is a fundamental requirement in Toyota plants to support flexibility and efficiency in one-piece flow cells, allowing teams to adjust staffing levels based on demand. Job rotation is a key feature that makes the tasks more **intrinsically motivating** (**job enrichment**).

Ringi Decision Making
While "Ringi" is a traditional Japanese method of management (bottom-up consultation), Toyota's documented principle is Nemawashi (consensus building). This is the process of making decisions slowly by consensus, thoroughly considering all options, before implementing rapidly. It involves discussing problems and potential solutions with all affected parties (including employees and external partners) to collect ideas and achieve agreement on a path forward.

The Foundation
The stability required to support the pillars and achieve the goals

Leveled Production (Heijunka)
Heijunka (Principle 4) is the process of leveling production by both volume and product mix to minimize inventory and keep the system stable. It is fundamental to eliminating unevenness (mura) and overburden (muri), allowing for standardized work.

Stable and standardized processus (Standardized Tasks)
Standardized Tasks (Principle 6) are the use of stable, repeatable methods everywhere to maintain predictability, regular timing, and regular output of your processes. Standardization is the foundation for flow and pull and is required before continuous improvements (Kaizen) can be made.

Visual Management (Visual Control)
Visual Control (Principle 7) is the use of simple visual indicators to help people determine immediately whether they are in a standard condition or deviating from it. This system makes problems visible—for example, Andon (alerting to defects) and Kanban (signaling consumption)—to ensure no problems are hidden.

Toyota Way Philosophy
The overall philosophy, rooted in Principle 1 (Base Your Management Decisions on a Long-Term Philosophy), provides the stable foundation and cultural DNA. The mission is to align the whole organization toward a common purpose that is bigger than making money and focus on adding value to customers and society.

© 2019 F. ROSIN

Toyota Business Practices (TBP)

The Toyota Business Practices (TBP) is Toyota's formal, eight-step methodology for continuous improvement and problem-solving, designed to translate the principles of the Toyota Way (such as long-term thinking and respect for people) into action.
The TBP process closely follows the classic Plan–Do–Check–Adjust (PDCA) cycle taught by W. Edwards Deming While TBP is often compared to generic 8D methods (for eight disciplines), TBP includes critical steps, particularly defining the ideal state, that most other problem-solving processes miss.

Here are the eight steps of the Toyota Business Practices (TBP), organized by the PDCA phases:

Plan Phase (Steps 1–5)

1. Clarify the Problem vs. the Ideal State
The starting point is to clarify the problem against the ideal state. Defining the ideal state is a critical step that avoids settling for incremental fixes to the current condition. The problem is explicitly defined as the gap between the target (standard) and the actual condition.

2. Break Down the Problem
The large problem (the gap defined in Step 1) must be broken down into smaller, more tractable pieces. This allows the team to set priorities and work toward challenging, yet achievable, goals.

3. Set the Target
For the pieces chosen to work on, the team establishes concrete, measurable targets. These targets serve as short-cycle, measurable guideposts along the improvement journey.

4. Root-Cause Analysis
The team must rigorously search deeply for the root cause of the gap between the actual condition and the target. This step often utilizes the "five whys" method: continually drilling down by asking "why" until the true source of the problem is identified.

5. Develop Countermeasures
Potential solutions, called countermeasures, are formulated. These solutions are considered provisional (hypotheses) until they are tested, based on the principle that "we do not really know what will happen until we try it".

Do Phase (Step 6)

6. See Countermeasures Through (Execute Plan)
This is the execution phase where the countermeasure (the proposed solution) is tried. This action is viewed as "running an experiment" to test the hypothesis formulated in the Plan stage.

Check Phase (Step 7)

7. Monitor Both Results and Processes
The team must closely monitor and analyze what occurred during the experiment. This step involves comparing the expected outcome with the actual outcome to learn what actually happened.

Adjust Phase (Step 8)

8. Standardize Successful Processes and Identify Gaps for Next Steps
If the countermeasure is effective, the practice is standardized. The system is kept in place until a better countermeasure is developed. This step also includes identifying further improvement projects, sharing the learning with others (yokoten), and starting the PDCA cycle anew.